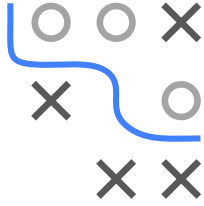
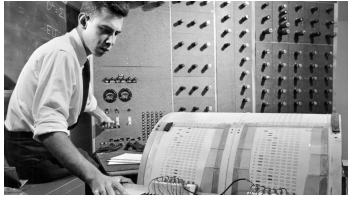
Introduction to Deep Learning



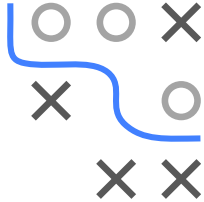
Introduction and Multilayer Perceptrons
Brief History



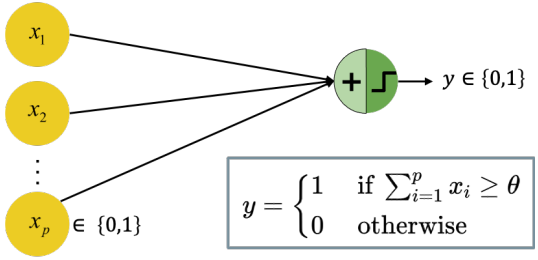
Learning goals

- Predecessors of modern (deep) neural networks
- History of DL as a field

A BRIEF HISTORY OF NEURAL NETWORKS I



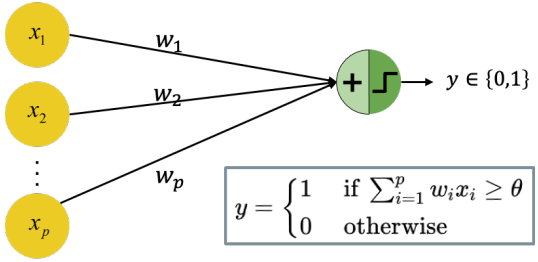
- **1943:** The first artificial neuron, the "Threshold Logic Unit (TLU)", was proposed by Warren McCulloch & Walter Pitts.

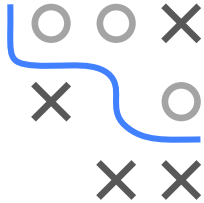


  - The model is limited to binary inputs.
  - It fires/outputs $+1$ if the input exceeds a certain threshold θ.
  - The weight are not adjustable, so learning could only be achieved by changing the threshold θ.

A BRIEF HISTORY OF NEURAL NETWORKS II

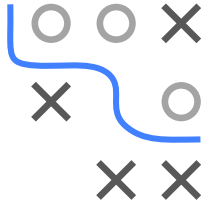
- **1957:** The perceptron was invented by Frank Rosenblatt.



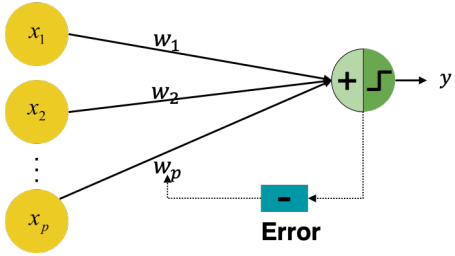


  - The inputs are not restricted to be binary.
  - The weights are adjustable and can be learned by learning algorithms.
  - As for the TLU, the threshold is adjustable and decision boundaries are linear.

A BRIEF HISTORY OF NEURAL NETWORKS III

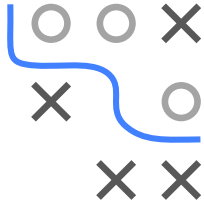


- **1960:** Adaptive Linear Neuron (ADALINE) was invented by Bernard Widrow & Ted Hoff; weights are now adjustable according to the weighted sum of the inputs.

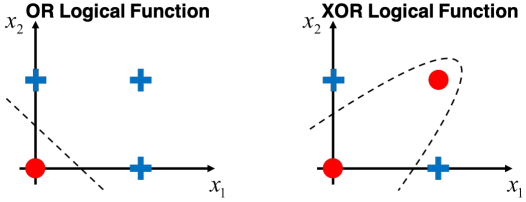


- **1965:** Group method of data handling (also known as polynomial neural networks) by Alexey Ivakhnenko. The first learning algorithm for supervised deep feedforward multilayer perceptrons.

A BRIEF HISTORY OF NEURAL NETWORKS IV

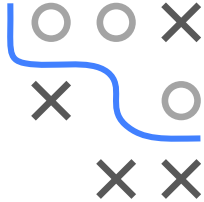


- **1969:** The first “AI Winter” kicked in.
  - Marvin Minsky & Seymour Papert proved that a perceptron cannot solve the XOR-Problem (linear separability).
  - Less funding $\Rightarrow$ Standstill in AI/DL research.



- **1985:** Multilayer perceptron with backpropagation by David Rumelhart, Geoffrey Hinton, and Ronald Williams.
  - Efficiently compute derivatives of composite functions.
  - Backpropagation was developed already in 1970 by Linnainmaa.

A BRIEF HISTORY OF NEURAL NETWORKS V

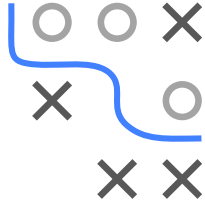


- **1985:** The second “AI Winter” kicked in.
  - Overly optimistic expectations concerning potential of AI/DL.
  - The phrase “AI” even reached a pseudoscience status.
  - Kernel machines and graphical models both achieved good results on many important tasks.
  - Some fundamental mathematical difficulties in modeling long sequences were identified.

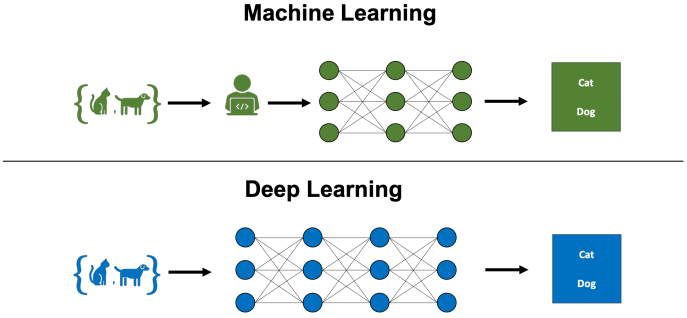


Figure: Symbolic image of an “AI Winter” (Fagella, 2019)

A BRIEF HISTORY OF NEURAL NETWORKS VI



- **2006:** Age of deep neural networks began.
  - Geoffrey Hinton showed that a deep belief network could be efficiently trained using *greedy layer-wise pretraining*.
  - This wave of research popularized the use of the term deep learning to emphasize that researchers were now able to train deeper neural networks than had been possible before.
  - At this time, deep neural networks outperformed competing AI systems based on other ML technologies as well as hand-designed functionality.



A BRIEF HISTORY OF NEURAL NETWORKS VII

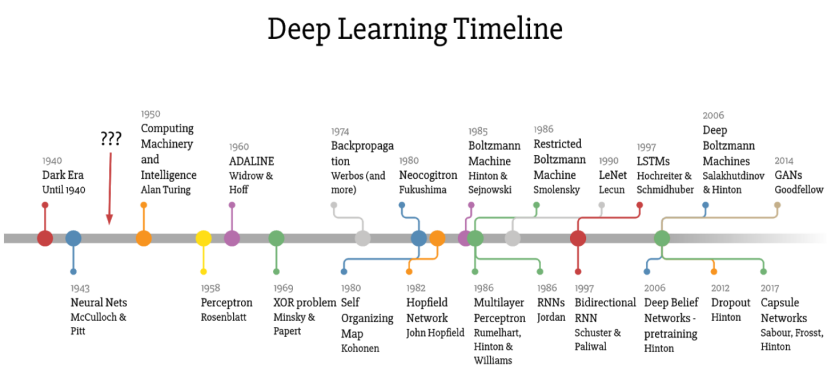
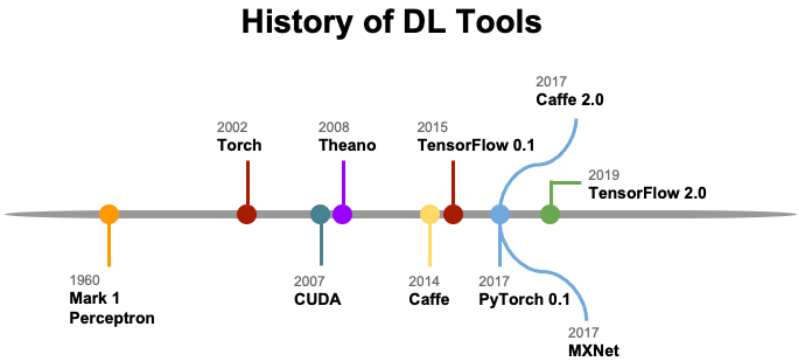
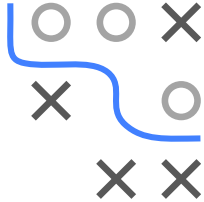


Figure: (Vazquez, 2018)

A BRIEF HISTORY OF NEURAL NETWORKS VIII

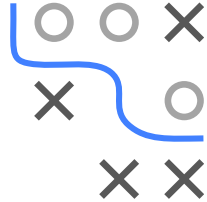




A BRIEF HISTORY OF NEURAL NETWORKS IX



Figure: IBM Supercomputer Watson (IBM, 2011)



- Watson is a question-answering system capable of answering questions posed in natural language, developed in IBM's DeepQA project.
- In 2011, Watson competed on *Jeopardy!* against champions Brad Rutter and Ken Jennings, winning the first place prize of $1 million.

A BRIEF HISTORY OF NEURAL NETWORKS X

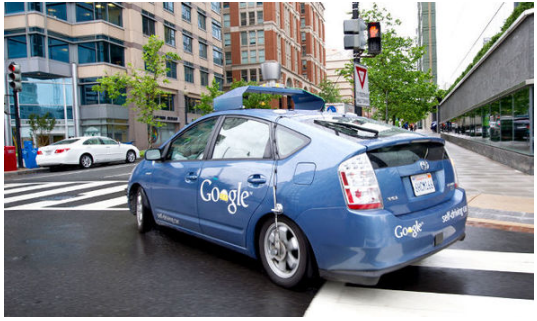
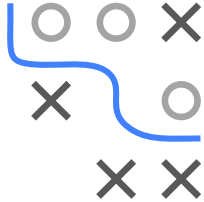
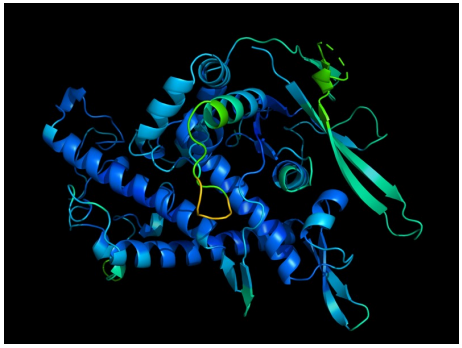


Figure: Google's self driving car (McCabe, 2016)



- Google's development of self-driving technology began on January 17, 2009, at the company's secretive X lab.
- By January 2020, 20 million miles of self-driving on public roads had been completed by Waymo.

A BRIEF HISTORY OF NEURAL NETWORKS XI



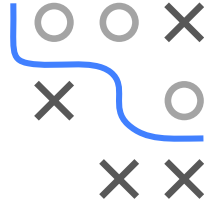


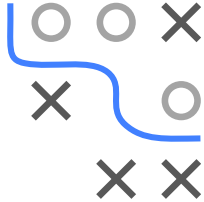
Figure: Protein structure by AlphaFold (DeepMind, 2022)

- **AlphaFold** is a deep learning system, developed by Google DeepMind, for determining a protein's 3D shape from its amino-acid sequence.
- In 2018 and 2020, AlphaFold placed first in the overall rankings of the Critical Assessment of Techniques for Protein Structure Prediction (CASP).

A BRIEF HISTORY OF NEURAL NETWORKS XII



Figure: AlphaGo (DeepMind, 2022)



- **AlphaGo**, originally developed by DeepMind, is a deep learning system that plays the board game Go. In 2017, the Master version of AlphaGo beat Ke Jie, the number one ranked player in the world at the time.
- While there are several extensions to AlphaGo (e.g., Master AlphaGo, AlphaGo Zero, AlphaZero, and MuZero), the main idea is the same: search for optimal moves based on knowledge acquired by machine learning.

A BRIEF HISTORY OF NEURAL NETWORKS XIII

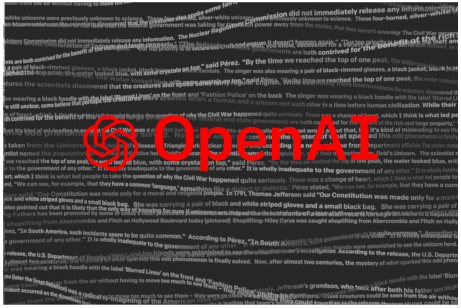
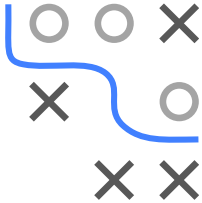
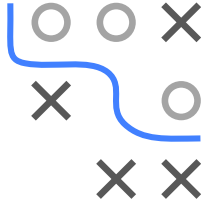


Figure: Logo OpenAI (OpenAI, 2023)



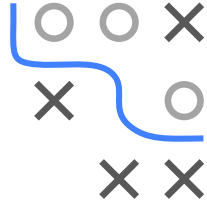
- **Generative Pre-trained Transformer 3 (GPT-3)** is the third generatation of the GPT model, introduced by OpenAI in May 2020, to produce human-like text.
- There are 175 billion parameters to be learned by the algorithm, but the quality of the generated text is so high that it is hardly possible to distinguish it from a human-written text.

WHAT THE FIELD LOOKS LIKE TODAY I



- A major shift in the 2020s was the move from highly specialized models to **foundation models**: large pretrained models that can be adapted to many downstream tasks.
- The frontier is increasingly **multimodal**: models can jointly process text, images, audio, video, and actions.
- Modern systems are also becoming more **interactive**: they can write code, search, retrieve documents, and call external tools.
- Examples of this broader landscape include **GPT-5** and **Claude Opus** for general-purpose assistants, **Stable Diffusion** for image generation, **DINOv3** for self-supervised visual representations, and **TabPFN** for tabular prediction.

WHAT THE FIELD LOOKS LIKE TODAY II



- Progress is now often driven by a combination of
  - better architectures and training objectives,
  - much larger datasets,
  - more compute and specialized hardware,
  - stronger software tooling and engineering.

  The modern DL pipeline is frequently: **pretrain at scale, then adapt with prompting, finetuning, retrieval, or tool use**.